

🔍 **Test Case A: Zero-State Base Model Verification (No Files Uploaded)**

- *Action:* Go straight to the query input bar without uploading files.
- *Input Question:* "What is the difference between a direct order and a non-directed order under SEC Rule 606?"
- *Expected Success Check:* The system will fall back to Gemini's native global knowledge base to correctly define how non-directed orders fall under Rule 606 public tracking metrics.

🔍 **Test Case B: Context Ingestion and RAG Validation**

- *Action:* Use the sidebar file uploader tool to drop your newly generated internal_routing_policy.pdf. Wait for the green database success message.
- *Input Question:* "Who does our firm route NMS options to according to our policy, and how much rebate do we receive from Apex?"
- *Expected Success Check:* The engine searches Chroma DB and answers correctly with: *Citadel Securities/Virtu Financial and a flat-rate rebate averaging \$0.0015 per share.*

🔍 **Test Case C: Contextual Compliance Cross-Referencing**

- *Action:* Provide the transaction log context directly into the chat prompt input bar.
- *Input Question:* "Review transaction TX-9988221 against our policy. Does routing 5000 AAPL shares to Apex match our active Rule 606 revenue rules?"
- *Expected Success Check:* The system extracts details from the JSON string, matches them to the indexed PDF document data, validates the rebate calculation (5000 shares * \$0.0015 = \$7.50), and confirms the transaction aligns with your structural rules.